Lab 6: Radiometer

Goal: Measure rotor speed at three illumination levels and check whether speed changes with time. Complete Pre-Lab 6, including Q6, before your section. Report: December 7, 11:59 PM.

1. Arrange the apparatus

1. Place the radiometer upright on a stable surface. Fix the camera on its tripod and frame the whole rotor. Keep the camera, bulb and background in the same positions for all runs.

2. Position the same lamp at the GTA-approved reference distance. Keep the lamp setting fixed. Place the lux meter beside the rotor at the same height, facing the lamp. Record distance [m] and illuminance [lux]; then remove the meter if it blocks the view.

2. Set the Chronos camera

Use the pendulum-lab recording baseline below. Open **Record Settings**; enter and record the settings before collecting data.

Control	Setting / action
Resolution	1280 x 1024 pixels.
Capture frame rate	1000 frames/s, the nominal pendulum-lab rate. Save the actual displayed value.
Analog / digital gain	0 dB (x1) / 0 dB (x1), as shown in the pendulum guide.
Exposure / shutter	The pendulum screenshot shows 342.64 microseconds at 2871.9 fps, not at 1000 fps. Use 342.64 microseconds as an initial trial setting if available; adjust only if needed for a sharp, visible vane edge, and record the final value.
Focus	Manually turn the lens focus ring until the vane edge is sharp in the live image. Focus has no universal numerical setting. Do not refocus between runs.
Brightness check	Keep vane edges visible and avoid saturated reflections. If too dark, ask the GTA to approve lighting/exposure changes before the test matrix; then keep them fixed.
Trigger / memory	Check the available record duration in seconds and the trigger window. Make sure the saved segment includes the illumination-on event and the required rotor passages.

After exposure/gain changes, perform the camera calibration required by the GTA. The final settings must remain unchanged across A1, B1, C1 and A2.

Collect four runs

3. Check one short recording

3. Record a short test and open Playback. Step through frames: the selected vane passage must be easy to identify. Watch one complete revolution and count **p**, the number of selected passage events per full revolution. Do not assume p = 4: if you follow only one identifiable vane, p = 1.

4. Choose a recording long enough to measure speed

4. Aim for **at least seven consecutive passage marks**, which give six time intervals. Record real elapsed time, not slow-motion playback duration. At 1000 capture fps, 10,000 recorded frames represent 10 seconds even if the exported movie plays much longer.

5. Check that the camera memory can hold the required segment **before** recording all conditions. For example, at 10 rpm with p = 4, six intervals take about 9 seconds; at 1 rpm they take about 90 seconds. These are examples, not expected measured speeds.

6. If the required duration exceeds memory, ask the GTA to reduce capture fps to an available rate that still resolves each passage, or approve a continuous known-rate recording. Record the actual rate and use it in MATLAB. Do not combine separated clips as if there were no gaps. Use the same supported observation window for the comparisons.

5. Record A1, B1, C1 and A2

7. A1: Begin from a documented initial state (rotor stopped / moving, cooling time). Start recording, switch on the lamp, and note the illumination-on frame in Playback. Collect the required consecutive passages.

8. Save the useful segment as **H.264** with **Text Overlay off**, as in the pendulum lab. Wait for saving to finish. Use a clear name such as Lab6_A1.mp4. Save a photo of Record Settings with the run.

9. B1 and C1: Use two other GTA-approved lamp distances. Measure the actual lux at each distance. Keep lamp power, camera settings and geometry otherwise unchanged. Let the apparatus return to a comparable initial state before each run; record cooling time.

10. A2: Return to the original A distance and repeat after comparable cooling. This checks repeatability. A warm return is a thermal-history run and must be labeled separately.

Record for every run: run ID, video filename, lamp setting, distance, lux, capture fps, exposure, both gains, cooling time, initial rotor state, p and illumination-on frame. Keep the raw video.

If time permits: Block the light after an illuminated run without touching the bulb. Continue recording and note when the light was blocked. Continued rotation may still be driven by remaining temperature differences.

Analyze and prepare the report

6. Mark passages and calculate speed

11. Unzip the **MATLAB files** bundle from the Lab 6 page. Run **Mark_Lab6_Passage_Frames.m** and select the movie. Left/Right move through frames; Up/Down change the step; Space marks; Backspace undoes; Q saves. Use a one-frame step at each crossing.

12. Mark the same event in the same direction, in increasing original frame order. Save the passage CSV. Run **Run_Lab6_Radiometer.m**, select that CSV, and enter the actual capture fps and your verified p. Check suspicious intervals against the movie.

Check one interval by hand: elapsed time = difference in frame numbers / capture fps. For consecutive passages, rpm = 60 / (p x elapsed time). For several intervals, rpm = 60 x number of intervals / (p x total elapsed time). The number of intervals is one less than the number of marks.

7. Compare the illumination runs

13. Fill **Lab6_Comparison_Log.csv**. Enter each passage CSV, capture fps, p, lux, distance, illumination-on frame and initial state. Time zero is illumination-on: t = (frame - illumination-on frame) / capture fps.

14. Choose the **same start and end times after illumination-on** for the runs being compared. Use only a window that the recordings actually cover and that contains enough verified passages. Enter that window in each log row. The code reports the actual first-to-last accepted passage coverage; it does not invent data at the edges.

15. Run **Compare_Lab6_Illumination_Runs.m** and select the log. Review interval rpm versus time, mean rpm versus lux, and the early/late summaries. Compare A1 with A2. Call a speed steady only if the time trace supports it.

8. Include these results in the report

16. Include one labeled setup photo and the camera settings; cumulative turns and interval rpm for one run; and mean rpm versus measured lux with A1/A2 identified. Explain early/late changes and any differences between the cooled repeats.

17. Use your required **Q6 molecular Monte Carlo** result to discuss how temperature contrast can affect gas momentum transfer. Relate the mechanism to the observed rotation. The molecular wall model is not a quantitative prediction of this bulb's rpm.

Use only relevant theory: rotor inertia x angular acceleration = gas torque - loss torque. A constant-torque, linear-loss model predicts an exponential approach to steady speed, but heating may change the gas torque during a run. Lux is not absorbed radiant power; internal pressure and vane temperatures are not measured.

Before leaving

Show the GTA one usable video, your recorded settings, verified p, one passage-speed result and the run-comparison plot. Save raw videos, passage CSVs, the completed log and exported figures for all group members.